

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12412472
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Nagpal; Varun et al.

Vehicle turn detection

Abstract

A turn detection system is configured to determine headings or a course of a vehicle over a period of time and evaluate whether the vehicle has registered a turn based on these headings/course. In some arrangements, upon detecting a turn, sensor data may be collected to determine one or more characteristics or attributes of the turn. Such data may indicate a loss event associated with the turn and be used to calculate a probability or risk of loss given the various characteristics of the turn. These probabilities may further be applied to determine various costs and premiums.

Inventors: Nagpal; Varun (Chicago, IL), Mukhtar; Yasir (Des Plaines, IL), Snyder; Jared S. (Chicago, IL), Walsh; Connor (Lake Forest, IL)

Applicant: ARITY INTERNATIONAL LIMITED (Belfast, IE)

Family ID: 58738123

Assignee: Arity International Limited (N/A, IE)

Appl. No.: 18/740468

Filed: June 11, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20240331534 A1	Oct. 03, 2024

Related U.S. Application Data

continuation parent-doc US 18121354 20230314 ABANDONED child-doc US 18740468
continuation parent-doc US 17007550 20200831 US 11605292 20230314 child-doc US 18121354
continuation parent-doc US 16166798 20181022 US 10769941 20200908 child-doc US 17007550
continuation parent-doc US 15873211 20180117 US 10140857 20181127 child-doc US 16166798

Publication Classification

Int. Cl.: **G08G1/056** (20060101); **G06Q40/08** (20120101); **G07C5/00** (20060101); **G08G1/00** (20060101); **G08G1/01** (20060101)

U.S. Cl.:

CPC **G08G1/056** (20130101); **G06Q40/08** (20130101); **G07C5/008** (20130101); **G08G1/0112** (20130101); **G08G1/0129** (20130101); **G08G1/20** (20130101);

Field of Classification Search

CPC: G08G (1/056); G08G (1/0112); G08G (1/0129); G08G (1/20); G06Q (40/08); G07C (5/008)

USPC: 705/4

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5390122	12/1994	Michaels	N/A	N/A
5462106	12/1994	Hanna	N/A	N/A
5469158	12/1994	Morita	340/988	G01C 21/28
6014610	12/1999	Judge	N/A	N/A
6028537	12/1999	Suman	N/A	N/A
6178368	12/2000	Olake	N/A	N/A
6502033	12/2001	Phuyal	701/472	G01C 21/30
8548739	12/2012	Lee	N/A	N/A
8694224	12/2013	Chundrlik, Jr.	N/A	N/A
8781669	12/2013	Teller	N/A	N/A
8930231	12/2014	Bowne	N/A	N/A
9127946	12/2014	Menon et al.	N/A	N/A
9147353	12/2014	Slusar	N/A	N/A
9176500	12/2014	Teller	N/A	N/A
9666067	12/2016	Nagpal	N/A	N/A
9905127	12/2017	Nagpal	N/A	N/A
10140857	12/2017	Nagpal	N/A	N/A
2004/0133346	12/2003	Bye	N/A	N/A
2004/0236477	12/2003	Chowdhary	N/A	N/A
2008/0004806	12/2007	Kimura	701/301	B60W 50/16
2009/0037052	12/2008	Ogasawara	701/41	G08G 1/167
2011/0066304	12/2010	Taylor	N/A	N/A
2011/0117903	12/2010	Bradley	N/A	N/A
2011/0125402	12/2010	Mitsugi	N/A	N/A
2014/0142786	12/2013	Huang	701/2	B60Q 1/40

2014/0257863	12/2013	Maastricht	N/A	N/A
2015/0006099	12/2014	Pham	N/A	N/A
2015/0112730	12/2014	Binion	705/4	G07C 5/008
2015/0153189	12/2014	Kim	701/400	G01C 21/3453
2016/0086285	12/2015	Jordan Peters	701/484	G08G 1/096827
2016/0171521	12/2015	Ramirez	701/409	G08G 1/096838
2016/0189323	12/2015	Wakabayashi	705/325	G06F 18/22

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
104504531	12/2014	CN	N/A
2004077283	12/2003	WO	N/A
2013107978	12/2012	WO	N/A

OTHER PUBLICATIONS

First Examination Report, IN Appln No. 201927011130, Dec. 2, 2020. cited by applicant
Self-Driving Cars and Insurance, Insurance Information Institute, Jul. 2016, downloaded from
<<http://www.iii.org/issue-update/self-driving-cars-and-insurance>> on Aug. 30, 2026. cited by applicant

International Search Report, PCT/US2017/049120, Sep. 29, 2017. cited by applicant

Extended European Search Report, EP 17847368, Mar. 18, 2020. cited by applicant

Office Action, CA 3036193, Apr. 28, 2020. cited by applicant

Primary Examiner: McNally; Kerri L

Attorney, Agent or Firm: Akerman LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 18/121,354, filed Mar. 14, 2023, which is a continuation of U.S. application Ser. No. 17/007,550, filed Aug. 31, 2020, and issued as U.S. Pat. No. 11,605,292 on Mar. 14, 2020, which is a continuation of U.S. application Ser. No. 16/166,798, filed Oct. 22, 2018, and issued as U.S. Pat. No. 10,769,941 on Sep. 8, 2020, which is a continuation of U.S. application Ser. No. 15/873,211, filed Jan. 17, 2018, and issued as U.S. Pat. No. 10,140,857 on Nov. 27, 2018, which is a continuation of U.S. application Ser. No. 15/496,336, filed Apr. 25, 2017, and issued as U.S. Pat. No. 9,905,127 on Feb. 27, 2018, which is a continuation of U.S. application Ser. No. 15/251,556, filed Aug. 30, 2016, and issued as U.S. Pat. No. 9,666,067 on May 30, 2017. The entire content of these applications is herein incorporated by reference in its entirety.

TECHNICAL FIELD

(1) Various aspects of the disclosure relate to detection of a vehicle turn. More specifically, aspects of the disclosure relate to determining whether a vehicle has made a turn and characteristics of the turn.

BACKGROUND

(2) Current navigation systems generally determine a vehicle's position and plot those positions against a predefined map. Accordingly, some existing navigation systems determine when a vehicle turns onto a new path/road based on the position of the vehicle rather than a determination of the vehicle's heading or course or any other characteristics of the turn. Moreover, in some instances, certain types of vehicle movements, or particular characteristics thereof, may have a high

correlation with other events such as accidents, traffic violations and the like. However, current systems do not generate such correlations nor are they configured in a predictive manner.

SUMMARY

(3) The following presents a simplified summary in order to provide a basic understanding of some aspects of the disclosure. The summary is not an extensive overview of the disclosure. It is neither intended to identify key or critical elements of the disclosure nor to delineate the scope of the disclosure. The following summary merely presents some concepts of the disclosure in a simplified form as a prelude to the description below.

(4) Aspects described herein provide a system, apparatus, method and/or computer-executable instructions for detecting vehicle movements such as turns in a real-time, on-demand or as-you-go manner. Additionally, various aspects provide the recordation or other determination of vehicle, driver and vehicle movement characteristics and information. Using such characteristics and information, a system may generate correlations between the vehicle movements and other events such as a risk of loss. These correlations, in aggregate across multiple vehicles and instances, may lead to the creation of a database, thereby providing a probability or risk of a particular event given a specified vehicle movement. Such correlations may be used in a variety of contexts including warning drivers of unsafe maneuvers, calculating insurance premiums (risks), peer monitoring of events and the like.

(5) Other features and advantages of the disclosure will be apparent from the additional description provided herein.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) A more complete understanding of the present invention and the advantages thereof may be acquired by referring to the following description in consideration of the accompanying drawings, in which like reference numbers indicate like features, and wherein:

(2) FIG. 1 illustrates a system for determining vehicle heading/course and detecting turns according to one or more aspects described herein;

(3) FIG. 2 illustrates an example computing device configured to detect turns, evaluate loss events and risk of loss according to one or more aspects described herein;

(4) FIG. 3 is a flowchart illustrating an example process for detecting a vehicle turn according to one or more aspects described herein;

(5) FIG. 4A illustrates the determination of a vehicle's position and a heading/course vector using the position information according to one or more aspects described herein;

(6) FIG. 4B illustrates a graph of vehicle headings or course according to one or more aspects described herein;

(7) FIG. 5 is a flowchart illustrating an example process for correlating risk with vehicle activity according to one or more aspects described herein; and

(8) FIG. 6 is a flowchart illustrating an example for determining insurance eligibility and premiums according to one or more aspects described herein.

DETAILED DESCRIPTION

(9) In the following description of the various embodiments, reference is made to the accompanying drawings, which form a part hereof, and in which is shown by way of illustration, various embodiments of the disclosure that may be practiced. It is to be understood that other embodiments may be utilized.

(10) Various aspects described herein may be embodied as a method, an apparatus, a system, or a computer-readable medium. Accordingly, those aspects may take the form of an entirely hardware embodiment, an entirely software embodiment or an embodiment combining software and

hardware aspects. Furthermore, such aspects may take the form of a computer program product stored by one or more computer-readable storage media having computer-readable program code, or instructions, embodied in or on the storage media. Any suitable computer-readable storage media may be utilized, including hard disks, CD-ROMs, optical storage devices, magnetic storage devices, and/or any combination thereof. In addition, various signals representing data or events as described herein may be transferred between a source and a destination in the form of electromagnetic waves traveling through signal-conducting media such as metal wires, optical fibers, and/or wireless transmission media (e.g., air and/or space).

(11) FIG. 1 illustrates a system in which various positioning systems may track a location and heading/course of a vehicle to determine whether the vehicle is executing or has executed a turn as well as to derive other driving characteristics. A course, as described herein, may generally relate to a single heading or a series of multiple headings. In one example, vehicle **101** may include one or more transceivers that are configured to communicate with a global positioning system (GPS) satellite **103**, which may include the receipt of positioning data. Through the positioning information received from satellite **103** through the receiver, a computing device of vehicle **101** may determine the location and heading/course of the vehicle **101**. In some instances, the vehicle computer (e.g., vehicle on-board computing device) may include pre-stored maps or maps obtained on-demand. The position information may then be used to graphically illustrate a position of the vehicle **101** within the maps (e.g., displaying a picture of a vehicle along road **109**). In some examples, the position information and associated map may be displayed on a display of a mobile device of a user, on a display within a vehicle and associated with an on-board vehicle computing device, and/or other display devices.

(12) According to other embodiments, a vehicle **101** or vehicle computing device may obtain positioning information through cellular or wireless information. In one example, the computing device or vehicle **101** may include a wireless or cellular transceiver configured to communicate with cellular or wireless access points such as access points **105**. The wireless or cellular transceiver may include a user's mobile device **107** that is linked to the vehicle's internal communication and computing systems. In some arrangements, a mobile device such as device **107** may be integrated as a hardware component of the vehicle (rather than a separable device of a user). Using a communication interface such as device **107**, vehicle **101** may perform triangulation calculations to determine the vehicle's position based on data received from access points **105** and/or from satellite **103**. Triangulation may include measuring the signal strengths between vehicle **101** (or other communication device associated therewith) and each of access points **105** and triangulating those strengths to identify a position of the vehicle **101**. In other arrangements, wireless local or wide area networks (WLAN or WAN's) may also be used to determine a vehicle's location. For example, if a vehicle is accessing a network through a Wi-Fi access point at a home **109** having a maximum range of 200 feet, the vehicle system **101** may determine the location of the Wi-Fi access point (e.g., by querying the access point or otherwise requesting location data) and subsequently identify its own location as within a radius of 200 feet of the location of home **109**. Access points may be provided at a variety of locations not limited to homes, communication towers, and/or satellites.

(13) GPS, cellular and other wireless network information may be used in combination to further refine or otherwise confirm positioning determinations of the vehicle **101** or of other devices. For example, in some areas, a GPS signal may be weak and thus provide a less granular determination of the location of vehicle **101**. In such an instance, cellular triangulation may be used to further refine the location determination to provide a more particular position of vehicle **101**. In some arrangements, the various types of positioning sources may be ranked such that if a first priority system (e.g., GPS) is unavailable or is providing information with a level of confidence lower than desired (e.g., signal is weaker than a threshold signal strength or resolution is lower than a threshold resolution), a device may go to a second priority system (e.g., cellular towers such as

access points **105**) and so on. In other examples, a turn detection system may increase a data collection rate from the first priority system if the first priority system is demonstrating weaker signal strength or is otherwise providing information with a lower level of confidence. The increased data collection rate may help to confirm or verify other pieces of data within the collection stream. Even when a turn detection system switches to a second priority system, the turn detection system may continue to monitor (e.g., collect data from) the first priority system to determine whether the resolution or strength of signal has improved. The ranking or priority may be designated by the turn detection system, a third party (e.g., an insurance company) and/or by the user. Data may be collected from multiple systems including both first and second priority location determination systems. The data may then be tagged with a reliability flag or attribute indicating whether the data from a particular system was collected with sufficient confidence.

(14) Beyond the devices described above that may be associated with vehicle **101**, vehicle **101** may further include one or more sensors (not shown) to sense vehicle or driving characteristics. Such sensors may include gyroscopes, accelerometers, cameras, motion detectors, light sensors, infrared sensors, impact sensors, thermometers and the like and/or combinations thereof. The vehicle **101** or other devices may use these sensors in a variety of manners, including to record images of vehicle events (e.g., traffic light status, turns, rapid acceleration, rapid deceleration, abnormal stops, swerving, hard braking, etc.), determine a severity of a crash (e.g., impact force, speed prior to impact, speed of deceleration), determine an external temperature of the area in which the vehicle is traveling, determine a time of day, determine presence and/or type of precipitation, determine a visibility (e.g., based on precipitation detectors and/or light detectors) and the like. Moreover, sensors of one vehicle may be used to help determine driving characteristics of another vehicle. For example, in the event of a crash of a first vehicle, the sensors, such as a camera, may be activated to help determine characteristics of that crash event, such as a traffic light status. The lack or disabling of sensors of the first vehicle may dictate or precipitate the activation and use of the sensors of the second other vehicle. In one arrangement, the turn detection system, upon detecting a turn event or loss event, may determine whether the vehicle includes various sensors. Different types of events may require different types of sensors. If not, the turn detection system may seek another vehicle in the vicinity (e.g., predefined proximity) with the requisite sensors, and activate or otherwise request data from those sensors. Aspects described herein may use or implement a variety of other configurations using such sensors.

(15) FIG. 2 illustrates a block diagram of a computing device (or system) **201** in a computer system **200** that may be used according to one or more illustrative embodiments of the disclosure. In one example, the computing device, sensors, positions systems and/or access points described in FIG. 1 may include components and configurations of device **201**. With continued reference to FIG. 2, the device **201** may have a processor **203** for controlling overall operation of the device **201** and its associated components, including RAM **205**, ROM **207**, input/output module **209**, and memory **215**. The computing device **201** may correspond to any of multiple systems or devices described herein, such as personal mobile devices, satellite, cellular or other wireless access points, ground vehicle-based computing devices including servers and home computing devices and other various devices. These various computing systems may be configured individually or in combination, as described herein, for determining a heading or course and/or position of a vehicle to detect turns (or other movements) of the vehicle. The systems may further collect vehicle and event data to determine and identify correlations between vehicle events or characteristics such as between certain types of vehicle movements and accidents.

(16) Input/Output (I/O) **209** may include a microphone, keypad, touch screen, and/or stylus through which a user of the computing device **201** may provide input, and may also include one or more of a speaker for providing audio output and a video display device for providing textual, audiovisual and/or graphical output. Software may be stored within memory **215** and/or storage to provide instructions to processor **203** for enabling device **201** to perform various actions. For example,

memory **215** may store software used by the device **201**, such as an operating system **217**, application programs **219**, and an associated internal database **221**. The various hardware memory units in memory **215** may include volatile and nonvolatile, removable and non-removable media implemented in any method or technology for storage of information such as computer-readable instructions, data structures, program modules or other data. Memory **215** also may include one or more physical persistent memory devices and/or one or more non-persistent memory devices. Memory **215** may include, but is not limited to, random access memory (RAM) **205**, read-only memory (ROM) **207**, electronically erasable programmable read-only memory (EEPROM), flash memory or other memory technology, CD-ROM, digital versatile disks (DVD) or other optical disk storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, or any other medium that can be used to store the desired information and that can be accessed by processor **203**.

(17) Processor **203** may include a single central processing unit (CPU), which may be a single-core or multi-core processor (e.g., dual-core, quad-core, etc.), or may include multiple CPUs.

Processor(s) **203** may have various bit sizes (e.g., 16-bit, 32-bit, 64-bit, 96-bit, 128-bit, etc.) and various processor speeds (ranging from 100 MHz to 5 GHz or faster). Processor(s) **203** and its associated components may allow the system **201** to execute a series of computer-readable instructions for, for example, processing sensor information, performing triangulation, calculating headings/courses of a vehicle, determining whether a turn has been made, performing network communications, calculating vehicle characteristics (speed, impact, rate of turn, deceleration, acceleration, etc.) and the like.

(18) A computing device such as device **201** (e.g., a personal mobile device, vehicle-based system, insurance server, etc.) may operate in a networked environment **200** supporting connections to one or more remote computers or devices (e.g., satellites, remote servers, drones, vehicles, personal devices). The network connections depicted in FIG. 2 include a local area network (LAN) **225** and a wide area network (WAN) **229**, and a wireless telecommunications network **233**, but may also include other networks. When used in a LAN networking environment, the computing device **201** may be connected to the LAN **225** through a network interface or adapter **223**. When used in a WAN networking environment, the device **201** may include a modem **227** or other means for establishing communications over the WAN **229**, such as network **231** (e.g., the Internet). When used in a wireless telecommunications network **233** (e.g., cellular or otherwise), the device **201** may include one or more transceivers (e.g., cellular, digital, etc.), digital signal processors, and additional circuitry and software for communicating with wireless computing devices **241** and **251** (e.g., mobile phones, portable customer computing devices, vehicle-based computing devices and systems, etc.) via one or more network devices **235** (e.g., base transceiver stations) in the wireless network **233**.

(19) Also illustrated in FIG. 2 is a security layer **260**, through which communications may be sent and managed between the device **201** and remote devices. The security layer **260** may comprise one or more separate computing devices, such as web servers, authentication servers, and/or various networking components (e.g., firewalls, routers, gateways, load balancers, etc.), having some or all of the elements described above with respect to the computing device **201**. In other examples, security layer **260** may correspond to software or hardware (e.g., encryption) within the device **201**. Encryption protocols may also be implemented through security layer **260**. Types of encryption may include public/private key pairs, Advanced Encryption Standard (AES) algorithms, Data Encryption Standard (DES) algorithms and other symmetric and asymmetric encryption protocols. Additionally or alternatively, security layer **260** may provide virtual private network (VPN) connections (e.g., using SSL or the like) through which certain types of data is sent to and received from other devices in the system.

(20) Although not shown in FIG. 2, various elements within memory **215** or other components in system **100**, may include one or more caches, for example, CPU caches used by the processing unit

203, page caches used by the operating system **217**, disk caches of a hard drive, and/or database caches used to cache content from database **221**. For embodiments including a CPU cache, the CPU cache may be used by one or more processors in the processing unit **203** to reduce memory latency and access time. In such examples, a processor **203** may retrieve data from or write data to the CPU cache rather than reading/writing to memory **215**, which may improve the speed of these operations. In some examples, a database cache may be created in which certain data from a database **221** (e.g., a database of property data, user data, user/owner interests or characteristics, etc.) is cached in a separate smaller database on an application server separate from the database server (e.g., at a personal mobile device, insurance processing device, or intermediary network device or cache device, etc.). For instance, in a multi-tiered application, a database cache on an application server can reduce data retrieval and data manipulation time by not needing to communicate over a network with a back-end database server.

(21) The network connections shown are illustrative and other means of establishing a communications link between the computers may be used. The existence of any of various network protocols such as TCP/IP, Ethernet, FTP, HTTP and the like, and of various wireless communication technologies such as GSM, CDMA, WiFi, and WiMAX, is presumed, and the various computing devices in home maintenance monitoring and reward system components described herein may be configured to communicate using any of these network protocols or technologies.

(22) In existing systems, GPS and other positioning systems have largely been used to provide navigation guidance. Such navigation systems simply recognize movements of the vehicle based on a point position of the vehicle and whether that position matches or maps to a predefined/anticipated route. For example, the navigation system will determine whether a vehicle has moved from a North-South road to an East-West road based on the point position changes of the vehicle. As such, existing navigation systems generally do not perform real-time, on-demand or as-you-go type detection/differentiation of a turn or other types of vehicle movements. However, as described herein, using this movement data (e.g., turn detection or differentiation between types of vehicle movements), a system may identify correlations between other types of attributes, characteristics and the like and the vehicle movements. For example, a system may identify correlations between risk of an accident or risk of loss based on the types of movement the vehicle makes, such as turns. Once such correlations are determined, a system may analyze a driver's behavior through vehicle movements to determine a risk of loss associated with that driver.

(23) Various methods may be used to perform real-time or on-demand detection of turns. One process includes determining the heading/course of the vehicle at predefined intervals for a specified period of time. FIG. 3 illustrates such an example process. In step **300**, for instance, a computing device, whether in the vehicle or remote from the vehicle, may initiate a sliding window as an evaluation period to determine whether a particular vehicle movement such as a turn has occurred. This initiation of a sliding window may be triggered by a variety of events such as the start of the ignition, a power-up event, user instruction, shifting to a particular gear (e.g., drive), reaching a particular speed threshold and the like and/or combinations thereof. The sliding window may be defined to be various lengths. In addition, the sliding window may be defined by a data collection rate. The data collection rate defines not only the amount of positions that are determined, but also the rate at which each position is determined. In one example, the sliding window is 13 seconds long with a data collection rate of 1 second (position is determined every second for 13 seconds). Based on empirical evidence, 13 seconds with a 1 second data collection rate provides an optimal time period for evaluating movements of the vehicle to detect turn events. Other periods of time or data collection rates may decrease the accuracy of detection applying the method discussed in FIG. 3, by either collecting too much data or too little data. However, other time periods and/or data collection rates may be used for the sliding window (less or greater than 13 seconds, less than or greater than 1 sec, etc.). In some instances, a system may compensate for

any increase in inaccuracy based on longer or shorter evaluation windows and/or longer or shorter data collection rates.

(24) In step **305**, the system may subsequently start to obtain position information for the vehicle according to the parameters of the sliding window, such as the data collection rate. As noted herein, the position information may be obtained from a variety of sources, including from GPS satellites, cell towers, Wi-Fi access points and the like. From the position information, the system may determine a heading/course of the vehicle in step **310** for each position determined for the vehicle. One method of determining the vehicle heading/course is to determine a vector between an immediately previous position and the currently obtained position. In some arrangements, other information from sensors such as gyroscopes and accelerometers may be used to supplement this determination of the heading/course. In a particular example, the amount of turning of the steering wheel may also be used to confirm or support a heading/course determination. For example, the steering wheel direction may be mapped to a heading/course of the vehicle or an amount of change in heading/course. Accordingly, such information may supplement a heading/course determination using GPS or triangulation. The heading/course may be defined in a variety of ways, including using magnetic north as a reference vector or using another predefined reference vector, and may be represented as an angle relative to the reference vector.

(25) In step **315**, a difference between each pair of immediately adjacent (time-wise) headings within the sliding window is calculated. The system may calculate two differences in heading/course, an unsigned difference (e.g., absolute value) as well as a signed difference. An unsigned difference may correspond to a sum of the absolute values of the headings, while a signed difference may correspond to a sum of the signed values of the headings. Once the differences have been determined for each pair of immediately adjacent headings within the sliding window, the system may then determine both an unsigned sum of the heading differences as well as a signed sum in step **320**. Using the unsigned sum, the system may determine whether a turn has occurred based on whether the sum exceeds a predefined threshold angle in step **325**. In one example, the threshold angle may be 58° . As such, if the unsigned sum is 58° or higher, the system may determine that the vehicle made a turn. On the other hand, if the unsigned sum is less than 58° , the system might not identify the movement as a turn. According to some empirical studies, use of a 58° threshold provides a high level of accuracy in determining whether a vehicle has registered a turn. Of course, other thresholds may be used. In some instances, if the unsigned sum is less than the turn threshold, but higher than a second threshold, the system may use other data to confirm whether the movement corresponds to a turn. Additionally, the system may further determine whether the turn is a right turn or a left turn based on the signed sum in step **330** if a turn is detected according to step **325**. For example, a negative signed sum may indicate a left turn while a positive signed sum may indicate a right turn. In some examples, if a turn is not detected, the process may return to step **315** to process a new window of vehicle movement data. Note the steps of FIG. 3 may be constantly performed irrespective of whether a turn has been detected.

(26) If the system detects a turn, the system may further record data in association with the turn in step **335**. The recorded data may include sensor information including speed, rate of turn, deceleration, acceleration, standard deviation of angle change, circular variance, amount or number of direction changes, accuracy of the location (e.g., accuracy of GPS or triangulation), moments of change, derivative of change in sign, whether an airbag deployed, whether a collision occurred, whether a turn signal was used, whether honking was detected and the like and/or combinations thereof. The system may constantly (e.g., continuously) or periodically collect sensor data irrespective of whether a turn is detected. However, this data might only be stored to long-term storage and/or in association with a turn event when a turn is detected. Additionally, only a portion of the collected data may be stored. For example, sensor data (or other data) for a predefined period within, overlapping or surrounding the turn event may be placed in long-term storage and/or associated with the turn event. Some systems may store a window of data that includes the time

window in which the turn was detected as well as a previous and subsequent period (e.g., 5 seconds, 10 seconds, 1 second, 1 minute, etc.). Alternatively, the data storage window may correspond to the turn event window as well as a subsequent period (e.g., 5 seconds, 30 seconds, 1 minute, 10 minutes, etc.). Other data may also be stored in association with the turn event, including driver characteristics (e.g., age, driving experience, accident history, etc.) and vehicle characteristics (e.g., make, model, year, safety features, engine type, horsepower, etc.).

(27) Turn detection may also be integrated into other vehicle functions such as speed control, turn signaling, headlight aiming and the like. In one example, upon detecting a turn, the system may employ speed control to cap the speed with which a vehicle may make a turn. Alternatively or additionally, the system may modify headlight direction based on the heading of the vehicle upon detection of a turn. Still further, a turn signal may automatically activate if a turn is detected. A variety of other controls may be integrated with the turn detection system and process.

(28) In one or more arrangements, data may be collected from sensors and other devices from other vehicles in a vicinity of the vehicle for which movement is being monitored and evaluated. In one example, when a turn is detected for a first vehicle, data from one or more second vehicles in a specified vicinity (e.g., a radius of 500 feet) may be queried for additional data, such as image/video/audio data, environmental information (temperature, weather, road slickness, humidity, etc.), traffic conditions, speed (of surrounding traffic or of the vehicle-in-question) and the like and/or combinations thereof. While in some systems, the other vehicles may collect driving/movement information at all times or otherwise irrespective of the turn event of the first vehicle, in other systems, the other vehicles may be triggered to begin recording data. Moreover, a system may request particular types of data from the other vehicles. Accordingly, the other vehicles might only activate sensors or otherwise record the data being requested. In systems where the vehicle might be collecting data irrespective of the detected movement event of the first vehicle, the other vehicles might only provide the requested data (e.g., types of data, periods of time for the data, etc.). In other instances, the other vehicles might provide all collected data for parsing/filtering by the requesting device or system. The data may be requested and sent to either a third-party system or to the vehicle (or device associated therewith).

(29) FIG. 4A illustrates the manner in which heading/course vectors may be determined based on position information. For example, each of points **401a**, **401b**, **401c**, **401d**, **401e** and **401f** may correspond to positions of a vehicle detected at 1 second intervals. To determine the vehicle's heading/course at a particular point such as point **401b**, the system may draw a vector **403a** between an immediately previous position **401a** and a current position **401b** and determine an angle of this vector **403a** relative to a reference line, such as magnetic north **405**. The determined angle may thus represent the heading/course of the vehicle at the corresponding point. For example, the heading/course at point **401b**, as identified by vector **403b**, may be determined to be 100 relative to magnetic north **405**. This process may be performed for each position **401c**, **401d**, **401e**, and **401f** within the sliding window to identify corresponding vectors **403b**, **403c**, **403d** and **403e** and headings/course, respectively.

(30) FIG. 4B illustrates how the vectors are then processed to evaluate whether the vehicle has registered a turn. As discussed, a difference between the headings/courses of each pair of vectors **403a**, **403b**, **403c**, **403d** and **40e** may be calculated. These differences may then be summed in both signed and unsigned manners to generate values for comparison to various thresholds and conditions in determining whether a turn has been made and a direction of the turn. In the illustrated example, the unsigned sum of the headings is 90° which is greater than the illustrative turn detection threshold of 58°, while the signed sum of the vehicle headings is -90°, indicating a left turn.

(31) The detection of vehicle movements such as turns may assist in developing a database of information relating to risks such as risk of loss in an insurance context. Accordingly, when evaluating an individual for insurance coverage, a system may consider various characteristics of a

user's driving behavior and compare these characteristics against the database associating risk of loss with various types of driving behavior or other characteristics.

(32) FIG. 5 is a flowchart illustrating a process for determining correlations between driving behavior and risk of loss. In step **500**, a system may collect a plurality of turn events registered for one or more vehicles. For example, an insurance system may identify multiple turns registered for multiple vehicles insured by that entity. In step **505**, the system may then filter the turn events down to those that have been associated with a loss event. For example, based on the collected data as discussed herein, the system may identify turns where an accident (or other loss event) occurred within the turning period or within a predefined amount of time of the turn event. Loss events may be detected using the sensor data collected from the turning vehicle or from one or more vehicles within a specified proximity. Moreover, loss events may be filtered based on a distance from the turning vehicle so that accidents or other loss events that are sufficiently remote from the turning vehicle are not associated. In one example, only sensor data collected for a particular area (e.g., 300 yard radius) around the vehicle may be used in identifying loss events (or other events). A time proximity may also be applied to filter events or data. Accordingly, only events occurring or data collected within a specified time period of a vehicle movement (e.g., a turn event) might be evaluated against the movement.

(33) In step **510**, the system may subsequently select a set of one or more characteristics or attributes to compare against the filtered set of turns associated with an accident. In some examples, a set of attributes may be pre-determined based on empirical studies showing that, for example, speed, turn rate, rate of acceleration, rate of deceleration, direction of turn and the like may be significant factors in whether a loss event occurs. In step **515**, the system may, for each attribute, determine a value or range of values of that attribute exhibiting at least a threshold percentage of loss events, e.g., a threshold percentage of the filtered population of turn events. In one example, the system may apply a threshold of 75% and determine a range of turn rates that encompasses 75% of the loss-associated turn events. In another example, the system may apply a threshold of 80% for determining a rate of deceleration (or range of rates) that covers 80% of the loss-associated turn events. The same or different probabilities/percentages may be applied for the different attributes. Alternatively or additionally, the system may determine a probability for a set of values of each attribute in step **520**. For example, a probability of a loss event may be determined for every 5 mph speed range (e.g. 1-5 mph, 6-10 mph, 11-15, mph, 16-20 mph, etc.) using the filtered data set of turn events. In another example, a probability of a loss event may be determined for multiple turn rate ranges. In a particular example, a first loss probability for turn rates 1-5 degrees per second may be determined based on the filtered turn data, while a second loss probability for turn rates of 6-10 degrees per second may be determined. These probabilities/risks may then be stored in step **525** for future lookup and use.

(34) Risk of loss may also be calculated based on non-movement attributes including driver attributes, such as age, driving experience, and driving infraction history, as well as geographic variables, such as location, population and the like. In one example, a risk of loss may be calculated for various age groups such as 18-21, 22-29, 30-35 and the like. Alternatively or additionally, risk of loss may also be calculated for different locations (e.g., cities, jurisdictions, zip codes, etc.). Additionally or alternatively, loss events or attributes contributing to a loss event might further be associated with a cost. These costs might be determined based on the cost of replacement, cost of repair or the like and/or combinations thereof. Such costs might be used in calculating a premium for a driver.

(35) Once risks of loss have been determined for the desired variables, a system may use this information in a variety of ways, including evaluating insurability of a vehicle or driver, calculating an overall risk for covering a particular vehicle or driver, determining an insurance premium, predicting accidents or other loss events and the like. FIG. 6 illustrates an example process by which loss probability information may be applied. In step **600**, for example, an insurance system

may receive information regarding a vehicle and a driver. The information may be received as part of an insurance quote request in one configuration. In step **605**, the system may determine one or more attributes of the vehicle and/or driver. In some examples, the system may retrieve driving behavior information of the driver based on public records or previously recorded driving data if, for instance, the driver or vehicle is currently insured by the same entity. Various other vehicle and driver attributes have been described throughout this disclosure. In step **610**, the system may determine a risk of loss associated with each of the one or more attributes. If multiple attributes are used, the system may derive a composite risk of loss in step **615** by combining the risks determined for each of the attributes. The risks may be combined by calculating a multiplicative product of the risks, averaging the risks (with or without weighting) and/or other algorithms. Based on the determined risk of loss or combined risk of loss, the system may then determine whether the driver or vehicle meets an insurability threshold in step **620**. If so, the system may further determine an insurance premium based on the level of risk associated with the vehicle and/or driver in step **625**. For example, the premium may increase linearly or multiplicatively based on the associated risk. If, on the other hand, the system determines that the driver or vehicle does not meet the insurability threshold, the system may respond to the driver with a notification of the same in step **630**. In some examples, the system may provide an explanation or details for the denial of coverage, such as the risk of loss being too high. The system may also provide detailed information including various driver characteristics or driver history that negatively impacted the determination.

(36) While the aspects described herein have been discussed with respect to specific examples including various modes of carrying out aspects of the disclosure, those skilled in the art will appreciate that there are numerous variations and permutations of the above-described systems and techniques that fall within the spirit and scope of the invention. For example, the processes, systems, apparatuses and other aspects of the present disclosure have been described with respect to insurance processing for buildings. However, the same or similar features may be applied to a variety of contexts including product sales, service offerings, real estate and the like.

Claims

1. A movement event detection apparatus comprising: a communication interface; a processor; and memory storing computer-readable instructions that, when executed by the processor, cause the movement event detection apparatus to: receive, from a remote device, position information for a vehicle through the communication interface, the remote device being exterior to the vehicle; determine headings of the vehicle based on the position information; collect a plurality of turn events associated with the vehicle based on the determined headings; filter the turn events down to those that are associated with a respective loss event, wherein the respective loss event occurred within a turning period that includes respective turn events or within a predefined amount of time of the respective turn events, and wherein the respective loss event is detected using sensor data collected from a respective vehicle; select a set of one or more attributes to compare against the filtered turn events associated with the respective loss event; determine a range of values of each respective attribute exhibiting at least a threshold percentage of loss events; and calculate an overall risk associated with the vehicle based on the determined ranges of values.
2. The movement event detection apparatus of claim 1, wherein the respective loss event is detected using sensor data collected from one or more vehicles within a specific proximity to the vehicle and within a time proximity to the respective loss event.
3. The movement event detection apparatus of claim 1, wherein the set of attributes includes at least one of a pre-determined speed, a turn rate, a rate of acceleration, a rate of deceleration, or a direction of turn.
4. The movement event detection apparatus of claim 1, wherein the movement event detection apparatus is further caused to: determine a probability for a set of values of each attribute.

5. The movement event detection apparatus of claim 1, wherein the movement event detection apparatus is further caused to: determine whether the vehicle registered a turn based on the determined headings; and employ vehicle functions upon determining that a turn event has occurred.
6. The movement event detection apparatus of claim 5, wherein the vehicle functions include capping a speed of the vehicle with which the vehicle can make a turn.
7. The movement event detection apparatus of claim 5, wherein the vehicle functions include modifying headlight directors based on a heading of the vehicle.
8. The movement event detection apparatus of claim 5, wherein the vehicle functions include automatically activating a turn signal when the turn is detected.
9. A non-transitory computer-readable medium storing computer-readable instructions that, when executed by a processor, cause a movement event detection apparatus to: receive, from a remote device, position information for a vehicle through a communication interface, the remote device being exterior to the vehicle; determine headings of the vehicle based on the position information; collect a plurality of turn events associated with the vehicle based on the determined headings; filter the turn events down to those that are associated with a respective loss event, wherein the respective loss event occurred within a turning period that includes respective turn events or within a predefined amount of time of the respective turn events, and wherein the respective loss event is detected using sensor data collected from a respective vehicle; select a set of one or more attributes to compare against the filtered set of turns associated with the respective loss event; determine range of values of each respective attribute exhibiting at least a threshold percentage of loss events; and calculate an overall risk associated with the vehicle based on the determined ranges of values.
10. The non-transitory computer-readable medium of claim 9, wherein the respective loss event is detected using sensor data collected from one or more vehicles within a specific proximity to the vehicle and within a time proximity to the respective loss event.
11. The non-transitory computer-readable medium of claim 9, wherein the set of attributes includes at least one of a pre-determined speed, a turn rate, a rate of acceleration, a rate of deceleration, or a direction of turn.
12. The non-transitory computer-readable medium of claim 9, wherein the movement event detection apparatus is further caused to: determine a probability for a set of values of each attribute.
13. The non-transitory computer-readable medium of claim 9, wherein movement event detection apparatus is further caused to: determine whether the vehicle registered a turn based on the determined headings; and employ vehicle functions upon determining that a turn event has occurred.
14. The non-transitory computer-readable medium of claim 13, wherein the vehicle functions include capping a speed of the vehicle with which the vehicle can make a turn.
15. The non-transitory computer-readable medium of claim 13, wherein the vehicle functions include modifying headlight directors based on a heading of the vehicle.
16. The non-transitory computer-readable medium of claim 13, wherein the vehicle functions include automatically activating a turn signal when the turn is detected.
17. A computer-implemented method comprising: receiving, from a remote device, position information for a vehicle through a communication interface, the remote device being exterior to the vehicle; determining headings of the vehicle based on the position information; collecting a plurality of turn events associated with the vehicle based on the determined headings; filtering the turn events down to those that are associated with a respective loss event, wherein the respective loss event occurred within a turning period that includes respective turn events or within a predefined amount of time of the respective turn events, and wherein the respective loss event is detected using sensor data collected from a respective vehicle; selecting a set of one or more attributes to compare against the filtered set of turns associated with the respective loss event; determining range of values of each respective attribute exhibiting at least a threshold percentage of

loss events; and calculating an overall risk associated with the vehicle based on the determined ranges of values.

18. The computer-implemented method of claim 17, wherein the respective loss event is detected using sensor data collected from one or more vehicles within a specific proximity to the vehicle and within a time proximity to the respective loss event.

19. The computer-implemented method of claim 17, wherein the set of attributes includes at least one of a pre-determined speed, a turn rate, a rate of acceleration, a rate of deceleration, or a direction of turn.

20. The computer-implemented method of claim 17, further comprising: determining a probability for a set of values of each attribute.
